

Scalability & Future Plan

Date	15 March 2024
Team ID	6aba3aa9ca93a27509644a39
Project Name	Comic craft AI
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dependence on external AI API	Story generation may be affected if the AI service is unavailable or slow	High
2	Limited handling of high numbers of concurrent users	Response time may increase when many users generate stories simultaneously	High
3	Limited persistent storage for generated comic content	Large amounts of generated stories and character data may become difficult to manage.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for a limited number of users	Deploy on scalable cloud infrastructure and use load balancing to support more concurrent users.
2	Data Storage	Limited/local storage	Integrate a scalable database and cloud storage for stories, characters, and generated content.
3	Performance	Suitable for normal usage	Introduce caching, asynchronous processing, and optimized API requests to improve response time.

4	Security	Basic API-key and environment configuration	Implement authentication, authorization, secure API-key management, input validation, and HTTPS.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User accounts and saved comic stories	1–2 Months	Users can securely save and manage their generated comics.
Phase 3	AI-generated comic images and character illustrations	3–4 Month	Makes the project more visual and improves the comic-creation experience.
Phase 4	Cloud deployment, scalable database, and multi-user support	5–6 Months	Allows Comic Craft AI to support a larger number of users and handle increased workloads.